EXPERIMENT No 14:

Programming a Wired 4WD Robot to Move Forward, Backward, Left, and Right Using Mixly and Push Buttons

Student Guide

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Objective:

In this experiment, you will learn how to program a 4WD robot to move forward, backward, left, and right using Mixly block programming software. Push buttons will be used to control the direction of the robot manually.

Materials Needed:

- Arduino Uno (or compatible microcontroller)
- L293D Motor Shield (for motor control)
- 4 DC Motors and Wheels (for robot movement)
- 4 Push Buttons (for controlling the directions)
- Jumper Wires (for making connections)
- Battery Pack (to power the motors)
- 4WD Chassis Kit (robot frame)
- Breadboard (optional for easier connections)

Project Overview:

You will connect 4 push buttons to the Arduino, each corresponding to a direction (Forward, Backward, Left, Right). When a button is pressed, the robot will move in the respective direction by controlling the motors with the L293D motor shield.

Steps to Build the Robot:

1. Assembling the Robot Chassis:

- Attach the 4 DC motors to the wheels and secure them onto the chassis.
- Make sure all wheels are properly aligned and rotate freely.

2. Connecting the Components:

- Motor Shield: Stack the L293D motor shield on top of the Arduino Uno.
- DC Motors:
  - M1 and M2: Connect to the left-side motors.
  - M3 and M4: Connect to the right-side motors.
- Push Buttons:
  - Connect one leg of each push button to digital pins on the Arduino:
    - Pin D2 → Forward button.
    - Pin D3 → Backward button.
    - Pin D4 → Left button.
    - Pin D5 → Right button.
  - Connect the other leg of each push button to GND using a resistor.
- Power Supply: Use a battery pack to power the robot through the motor shield.

3. Circuit Diagram:

Ensure all connections are made correctly:

- Motor shield → Arduino Uno.
- Motors → Motor shield.
- Push buttons → Digital pins (D2, D3, D4, D5) and GND.

Programming the Robot Using Mixly:

1. Open Mixly: Launch the Mixly software on your computer.
2. Create the Program:
    - Start by adding the Button blocks for each of the push buttons (Forward, Backward, Left, and Right).
    - Use IF-ELSE statements to check which button is pressed.
    - Based on the button press, use Motor Control blocks to drive the robot's motors in the desired direction.

Basic Program Outline in Mixly:

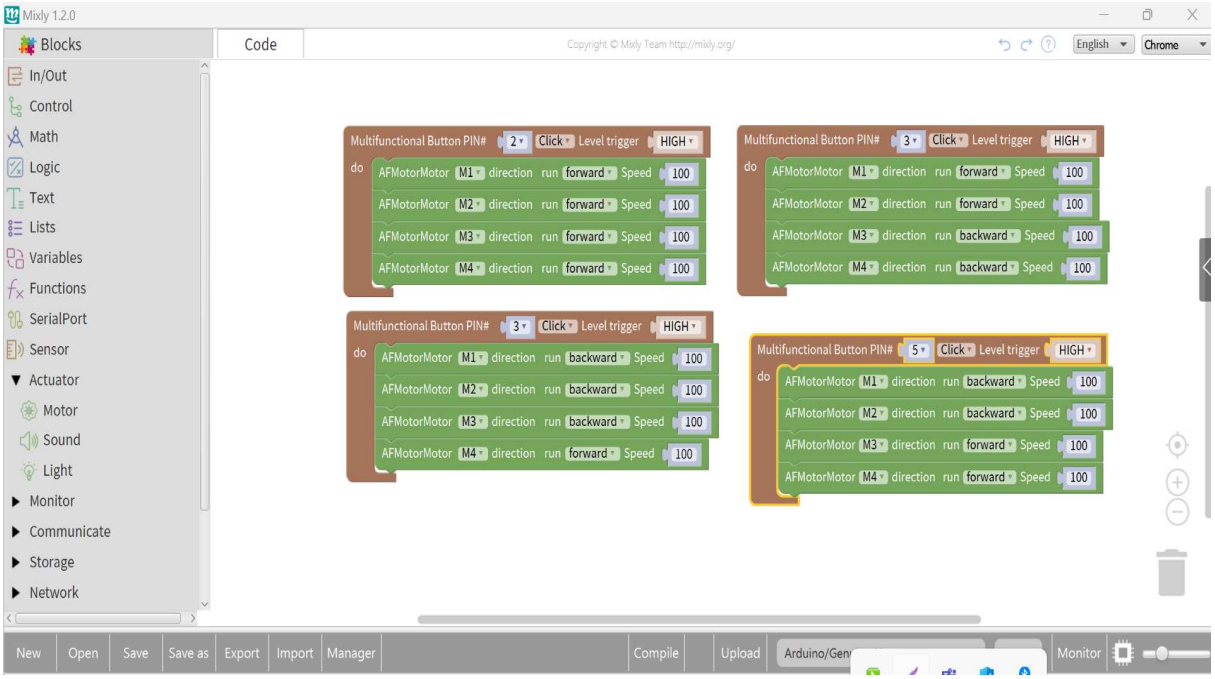
- Forward Movement: When the forward button (D2) is pressed, both left and right motors should move forward.
- Backward Movement: When the backward button (D3) is pressed, both left and right motors should move backward.
- Left Turn: When the left button (D4) is pressed, the right motors should move forward and the left motors should stop.
- Right Turn: When the right button (D5) is pressed, the left motors should move forward and the right motors should stop.

Mixly Block Diagram:

Here is a simplified outline of how the Mixly blocks will be arranged:

- Start Block.
- Button Control for each of the 4 directions.
- IF-ELSE:
    - If Forward button is pressed, move motors Forward.
    - If Backward button is pressed, move motors Backward.

- If Left button is pressed, move motors to turn Left.
- If Right button is pressed, move motors to turn Right.



Testing the Robot:

1. Upload the Code: Connect your Arduino to your computer and upload the program from Mixly.
2. Power the Robot: Attach the battery pack to the motor shield to power the motors.
3. Test Each Button:
   - Press the Forward button and check if the robot moves forward.
   - Press the Backward button to move the robot backward.
   - Test the Left and Right buttons to turn the robot.

Code Explanation:

- Button Press Detection: Each button is linked to a specific digital pin on the Arduino. When pressed, the corresponding pin will read as HIGH, triggering the movement.
- Motor Control: The motor shield controls the direction of the motors. By sending different signals to the motors (forward or backward), the robot moves accordingly.

Project Enhancements:

- Speed Control: Add a potentiometer to control the speed of the robot.
- Obstacle Detection: Integrate an ultrasonic sensor for obstacle avoidance, making the robot more autonomous.
- Multiple Functions: Combine manual button control with autonomous navigation modes.

Reflection Questions:

1. How does each push button control the direction of the robot?
2. How would you modify the program to make the robot stop automatically after moving for a certain time?
3. What other sensors could you add to make the robot more interactive?

Project Outcome:

By completing this project, you will have programmed a 4WD robot to respond to button presses and move in various directions. This hands-on activity teaches you about motor control, wiring circuits, and using Mixly for programming.

This handout provides step-by-step guidance for building a manually controlled 4WD robot using push buttons and Mixly. Keep experimenting with the code and add more functionalities to enhance your robot!